

Evidence-Based Biomaterial Dataset for EEG/ECG/EMG Electrodes

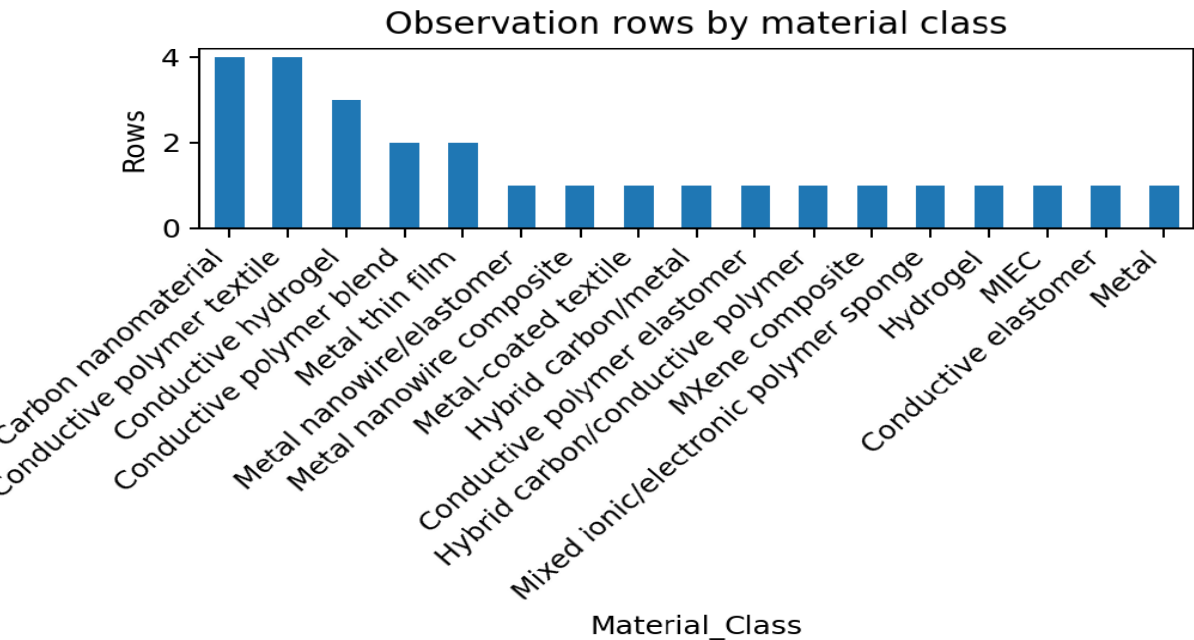
Research report and dataset specification — 2 September 2026

Executive summary.

The accompanying workbook contains 27 traceable observation/configuration rows from 19 primary research articles. The dataset separates material composition from architecture and interface design, retains raw source values, records measurement conditions where available, and distinguishes measured outcomes from derived variables. The evidence does not support a universal best electrode; instead it supports context-conditioned Pareto optimization across signal quality, impedance, adhesion, mechanical compatibility, biological response, durability and practicality.

Search status.

This is a high-quality evidence seed assembled by targeted searches of PubMed/PMC and publisher-hosted primary research pages. It is deliberately conservative. It is not a PRISMA-complete systematic review, and some articles exposed only abstracts/snippets during this pass. Such cases were not padded with inferred numerical values.



nt seed dataset by material class; not a literature-prevalence estimate.

Figure 1. Current seed dataset by material class; not a literature-prevalence estimate.

1. Methodology and inclusion criteria

- Prioritize peer-reviewed primary experimental research involving EEG, ECG, EMG or general biopotential electrode acquisition.
- Extract numerical values only when explicitly visible in the primary source; preserve the raw value and source location.
- Separate material, architecture, substrate, interface/adhesive layer and experimental condition.
- Keep comparative controls such as Ag/AgCl and commercial dry electrodes where the source provides them.
- Use reviews for taxonomy/context only; do not replace primary quantitative evidence with review numbers.

Key exclusion: unverified values, invented cost estimates, or numbers inferred from qualitative statements are excluded from the main quantitative dataset.

2. Taxonomy and interface design

Material_Name	Material_Class	Material_Subclass	Typical_Architecture	Design_Rationale
Ag/AgCl	Metal/metal salt	Clinical wet reference	Wet	Strong comparator; dehydration limits long wear
Graphene	Carbon nanomaterial	Graphene film/textile/paper	Dry flexible	Electrical performance strongly affected by architecture and
AgNW/PDMS	Metal nanowire/elastomer	AgNW composite	Dry flexible	Conductive and compliant; adhesion can be tuned through
PEDOT:PSS	Conductive polymer	Textile/elastomer/hydrogel	Dry or wet	Formulation changes conductivity, adhesion and mechanics
Ti thin film	Metal thin film	Architected/porous	Dry	Microstructure affects impedance and sEMG
MXene Ti3C2Tx	2D material	Cellulose aerogel composite	Dry high-density	High-density EEG with gel-comparable signal similarity
Gelatin	Natural polymer	PEDOT:PSS hydrogel	Wet hydrogel	Self-adherent, biodegradable matrix
PVA/CMC	Polymer blend	PEDOT:PSS hydrogel	Wet hydrogel	High water content and tissue-like interface
PVA/PVP/PDA	Hydrogel	PDA nanoparticle hydrogel	Wet hydrogel	Self-adhesive EEG platform
PPy-PCNF/PAA/Gel	Conductive hydrogel	PPy/cellulose nanofiber hydrogel	Wet adhesive hydrogel	High stretchability and adhesion; recent 2026 evidence

3. Key literature-derived quantitative observations

System	Primary evidence	Key measured result	Implication
Graphene textile	Human ECG	281.8 k Ω at 10 Hz; 24.8 k Ω at 1 kHz; rest SNR 31.6 dB; walking SNR 20.2 dB	Strong ECG fidelity despite higher contact impedance; textile/contact interface matters.
AgNW/PDMS	Human ECG + MTT	25.4 dB SNR vs 24.6 dB Ag/AgCl; ~90% viability at 7 days	Strong dry-electrode candidate with both signal and biological evidence.
Adhesive AgNW/PDMS	Human adhesion/ECG	1.09 N max skin adhesion; $\geq$ 24 h tested wear; 0.6 mg/cm ² AgNW	Architecture and PDMS cure ratio are decisive interface variables.
PEDOT:PSS/PU	Human EMG	18.2 $\pm$ 0.3 dB SNR vs 15.7 $\pm$ 0.3 dB wet Ag/AgCl; 99.57% classification	Shows dry polymeric interfaces can match/exceed wet control on a system level.
PTG graphene/PEDOT:PSS	Human EEG/ECG/EMG	24 Ω /sq; 4142 S/cm; ~32 k Ω at 100 Hz	Strong combined electrical/conformal profile; system-level scaling relevant.
MXene MXtrode	Human EEG	5.15 k Ω ·cm ² vs 1.21 k Ω ·cm ² Ag/AgCl; ERP $\rho \geq$ 0.95	Higher normalized impedance does not preclude high EEG similarity.
Reversible sponge	Human biopotential	3.5 k Ω contact impedance; 9.82 dB vs 7.92 dB control; 10,000 cycles	Reversibility can be optimized independently of peak SNR.
Silicone dry electrode	14 humans	39.43 k Ω ; 1.51 $\pm$ 0.10 MPa modulus; 99.20 $\pm$ 2.50% R-peak recall; 14.06 $\pm$ 0.08 dB	Human validation with explicit sample size.
Gold pin dry electrode	Human EEG	~300 k Ω at 100 Hz; SNR 1.27 vs 1.30 dB Ag/AgCl	Geometry matters; SNR scale is protocol-specific.
PPy-PCNF/PAA/Gel	Human ECG	2.26 S/m; >700% stretchability; 21.8 kPa adhesion	Integrated hydrogel design is promising; accessible abstract lacked.

These values must not be pooled as if measured under a common protocol. In particular, impedance may be normalized by area or measured at different frequencies and pressures.

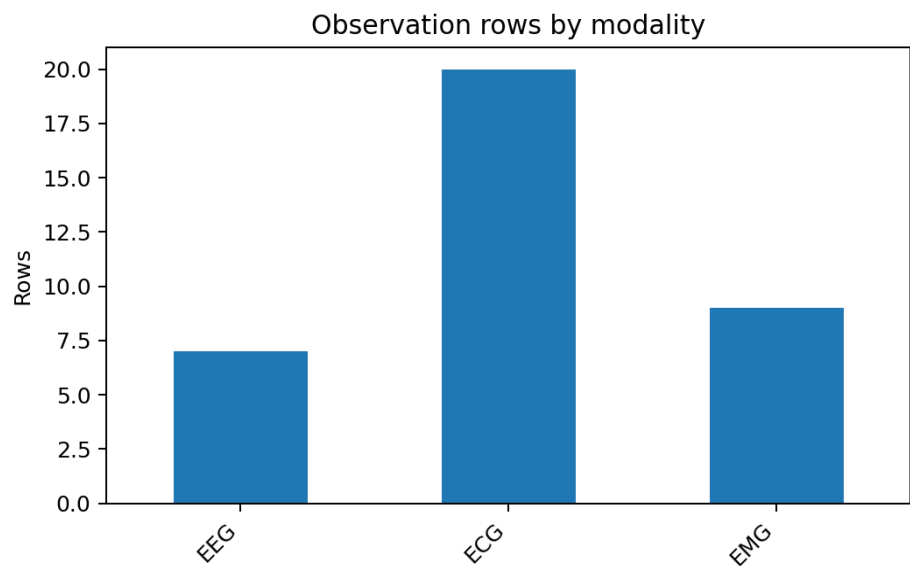


Figure 2. Modality distribution in the current seed.

4. Variable rationale

Variable group	Why it matters	Modeling recommendation
Material composition	Controls electronic/ionic transport, electrochemistry, hydration and biological response.	Use composition-aware descriptors; do not let material class stand in for composition.
Architecture/geometry	Controls real contact area, local pressure, hair penetration, conformability and motion coupling.	Encode geometry, substrate and multilayer structure explicitly.
Impedance spectrum	Interface impedance is frequency-dependent and sensitive to skin condition.	Store frequency with every value; consider spectral features.
Conductivity	Bulk conductivity does not directly equal skin-electrode interface performance.	Keep bulk conductivity separate from interface impedance.
Adhesion	Stable contact reduces micro-slippage and motion artifacts.	Keep force, duration, sweat/water and removal behavior separate.
Mechanical compatibility	Modulus mismatch and fatigue affect comfort and contact stability.	Model modulus, stretchability, cycling and hydration together.
Biocompatibility	Long-term wear requires low irritation/inflammation and acceptable cell/tissue response.	Keep assay type, cell line and duration with every value.
Signal performance	Complete-system validation is the most application-relevant outcome.	Model by modality and protocol; avoid cross-protocol ranking.

5. Multi-objective optimization

For use case c , define direction-corrected standardized objectives $z_i(x,c)$ and optimize a vector. A deployment-specific utility can be $U(x|c)=\sum w_i(c)z_i(x,c)$, with weights summing to 1. The weights should be chosen from the intended application rather than from reporting frequency. In parallel, compute the Pareto front across signal fidelity/SNR, interface impedance and stability, adhesion/motion robustness, mechanical compatibility, skin compatibility, hydration/drying resistance, durability/reuse and manufacturability.

Recommended targets: modality-specific SNR/signal fidelity, frequency-indexed impedance, adhesion, mechanical compatibility, biocompatibility and stability. An overall index should be a decision output, not an input feature when constructed from those same outcomes.

6. ML development recommendations

- Use study/paper-grouped cross-validation. Random row splitting can leak repeated configurations from the same paper.
- Start with Random Forest, gradient boosting/XGBoost and regularized multi-output models; use Gaussian Processes when the cleaned dataset remains small and uncertainty is important.
- Defer deep neural networks until the number of independent studies/configurations is much larger.
- Model measurement context explicitly: frequency, electrode area, pressure, skin preparation, hydration, hair, sweat, motion, amplifier and filtering.
- Use Pareto-front analysis before imposing application-specific weights.
- Track missingness and publication bias; NR is not zero.

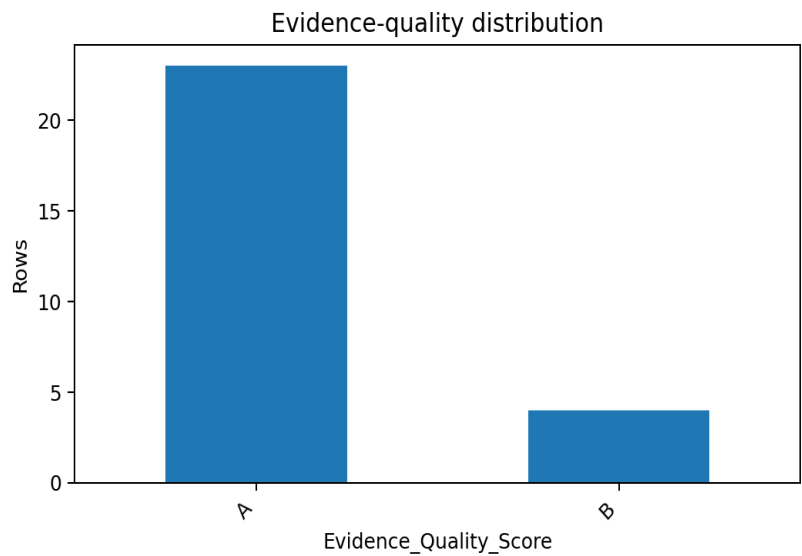


Figure 3. Evidence-grade distribution in the current seed.

7. Limitations and uncertainty

- The literature is methodologically heterogeneous. Impedance, SNR and adhesion are not directly comparable without their test conditions.
- Many hydrogel papers report strong conductivity/adhesion and qualitative ECG/EEG/EMG validation but omit standardized signal metrics.
- Several studies use small human samples or single-subject demonstrations.
- Material-level properties and full electrode-system properties must not be conflated.
- Novel-material publication bias is likely; negative/failed designs are less visible.
- Recent 2026 studies have limited replication evidence.
- This dataset is a conservative evidence seed; production ML should expand full-text and supplementary extraction and independently verify every numerical observation.

8. Evidence-based conclusions

- Ag/AgCl remains a strong clinical reference, but the literature does not justify calling it universally optimal for long-duration wearable use.
- PEDOT:PSS is a particularly versatile platform because it can be engineered as textile, elastomer, dry film or hydrogel; formulation and architecture materially change performance.
- Graphene/AgNW systems offer strong electrical performance with flexible architectures, while adhesion and sweat/motion coupling determine real-world robustness.
- MXene EEG arrays show why interface impedance alone is an insufficient ranking target.
- Conductive hydrogels provide an attractive route to soft, hydrated, self-adhering interfaces, but durability and conductivity depend strongly on formulation.
- The scientifically defensible final output is a context-conditioned Pareto set, not a universal material ranking.

9. References

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